Evaluating Tangle as an alternative to blockchain for service composition in IoTs

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***Abstract*—In today's world, Internet of Things (IoTs) have become prevalent and produce a large amount of data. Many researchers and industries have explored ways to improve accessibility through compositions of IoTs services. However, many research studies have already outlined security and privacy related issues in IoT. The inception of blockchain technology and its adaptation have given rise to distributed ledger technologies (DLT). Many industries and researchers have proposed solutions to utilize blockchain technology in IoT for their collaborative service. However, due to blockchain architecture and its limitations, it is not suitable for IoTs in a long term.**

To overcome some of the blockchain limitations, a new technology namely IOTA Tangle was proposed. Tangle is a based-on Directed Acyclic Graph (DAG) architecture rather than sequential blockchain. As Tangle is based on DAG, it can process multiple requests from multiple clients concurrently, and therefore, its architecture is promising for service composition. With this paper, the authors highlight that how DAG-based DLT technology Tangle can be used for service composition schemes in IoTs using a case study.

***Index Terms*—Distributed Ledger Technology (DLT), Blockchain, IOTA Tangle, Directed Acyclic Graph (DAG), Public key infrastructure (PKI), Internet of Things (IoT), Service Composition**

I. Introduction

In today's world, the extensive use of the Internet for exchanging information had kindled the idea of the Internet of Things (IoT). Any electronic device connected to the internet wirelessly to provide innovative solutions and accessibility can be called an IoT device. IoTs have become so prevalent in today's world that they can be seen in our day-to-day life including homes, offices, hospitals, government buildings, roads, and traffic signals. Some examples of IoTs are mobile, wearable, sensors, surveillance cameras, medical devices, ultrasound sensors, RFID devices, etc. Many researchers have already outlined that IoTs has disrupted many industries [1][2]. Many in-depth surveys and researches [3][4] have been published on IoTs, their limitations, and applications in many varieties of fields. According to the articles [5][6], there are currently 50 billion connected IoTs as of 2020 and is expected to grow exponentially in the near future.

On the other hand, service orientated architecture (SoA) has been the building block for today's modern enterprise applications in which business workflows are divided into smaller services and their composition are used to serve business needs. With the ever-increasing number of IoTs, many researchers have applied the composition of IoT services in various fields and domains [7][8]. The authors Hamzei at el.[9] have eloquently explained the service composition techniques in IoTs. As IoTs are seen as numerous network devices connected over the Internet, the composition of multiple IoT services becomes viable and important. However, many research studies have underlined the security and privacy issues in IoTs [10][11]. Also, due to IoT's heterogeneous[12] nature, evaluation and verification of data received from IoTs still have to be administered with centralized trusted entities [13].

With the success of blockchain technology [14] and its immutability and inherent security, many researchers have explored its applicability in different domains including IoT [15][16]. However, over the years many researchers have emphasized the limitation and challenges in blockchain-based technologies such as scalability, network, power consumption, and higher computation power to name a few [17][18][19].

The continuous research on overcoming the challenges in blockchain, a newer technology IOTA Tangle [20] designed for low compute devices was proposed. Tangle is based on Direct Acyclic Graph (DAG) rather than blockchain. It is a lightweight structure capable of running on IoT. Furthermore, tangle provides greater flexibility to scale the network and transactions. As tangle structure is based on DAG, it can process and store multiple transactions from multiple IoTs concurrently. Hence, its structure and architecture is befit for service composition. Therefore, through this paper, we highlight that how a service composition scheme can be implemented using tangle for IoTs.

Contributions: The main contributions of this study are-

- A new tangle architecture for IoTs.
- Service composition in IoTs using tangle
- Secured data sharing using tangle

Structure: The rest of the paper is organized as follows: The section II will provide the background on IoTs, service composition, and its applicability in IoTs. Also, the section will outline the use of blockchain in IoTs, service composition using blockchain, and tangle. The core components of the proposed smart home model are described in section III. In section IV, we propose and implement a smart home use-case model based on Tangle in service composition scheme. The results and evaluation of our model are described in section V. Section VI includes some considerations of implemented use-case model and security analysis. Section VII will describe any future work on the current approach. And finally, the section VIII will provide the conclusion of the research work.

II. Background

IoTs are small electronic devices connected to the internet to facilitate and aid accessibility in our day-to-day life. The adaptation of IoTs has grown extensively in the past few years and is projected to grow exponentially [6], thanks to the expansion of the internet to every corner of the world and the reducing cost of manufacturing and buying IoTs [21]. In today's world, IoTs have become so prevalent that one or more IoTs can be seen in every household, office, hospital, and government buildings. According to authors Islam et al. [22], IoTs will be frontier in the future and will disrupt many industries. As these IoTs produce a huge amount of data, many research studies have been performed on utilizing these IoTs and their services [23] to improve human surroundings for a better life and the world.

A. Service composition in IoTs

The authors Hamzei et al. [9] has very eloquently explained and summarized the composition of IoTs services and the service composition techniques in IoTs. The authors have divided techniques into 4 categories of work one of which is Service Oriented Architecture (SoA) which has been the industry defacto for developing software and application components to be used by any other components for many years. similarly, service composition has been distinctly described in [23] with an expressive example.

Before we delve further, one should be aware of what is "service" and what "service composition" means. A service is a software functionality that enables access to more capabilities and is offered through an interface [24]. In simple terms, it is a way to provide one device capabilities and offer to other devices for use. "Service composition" is a way of composing different devices' capabilities into one service and offering that capability to different devices as a service. Figure 1 illustrates an example of service composition.

In the figure 1, service s_a takes data from three different services namely s_1, s_2 and s_3. Then, it composes the data received from these services, makes a further observation, and sends it to services s_x and s_y for further composition and observation. To illustrate it with a real-world example, let's take an example of a smart home. In a smart home, a smart light bulb can request or receive data from a smart motion sensor to turn ON or OFF based on any activity recorded by a motion sensor (when someone enters or leaves the room).

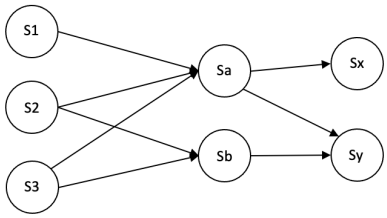


Fig. 1: An illustrative example of service composition

B. Blockchain in IoTs

Even though IoT and their service compositions are promising for building and shaping the future of connected devices, services and accessibility, many research studies have underlined the security and privacy issues associated with IoT [10]. With the adaption of blockchain technology due to its immutability and inherent security, many researchers have proposed solutions based on blockchain that solves some of the existing IoTs concerns [25].

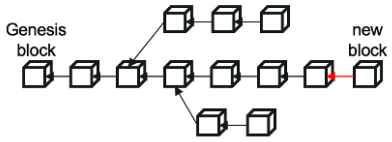


Fig. 2: A sample blockchain two two other dishonest branches

A blockchain is an immutable decentralized distributed ledger or store which is a chain of a block consisting of transaction records. In a blockchain, a block consisting of several transactions linked to previous transactions block by a hash pointer. This architecture makes it impossible to modify any previous transaction or block because to modify a previous transaction or block, one need to modify (change reference) all other block attached to that block and this require immense computing power greater than 50% [26][27] of a blockchain network. Figure 2 layouts a sample blockchain with two dishonest branches trying to overpass the honest branch.

According to authors Ali et al. [13], security and privacy are still major issues with IoTs and IoTs infrastructure is susceptible to weakness and attack. The authors did a comprehensive study on applications of blockchain in IoTs and highlighted some of the key challenges with IoT. Additionally, authors have emphasized the centralized nature of IoT infrastructure, noting it as a single point of failure, susceptibility to DDoS attack, and potentially not enough for exponentially growing

IoTs in the future. The authors have outlined the benefits of blockchain as a decentralized and secure immutable data store.

However, the limitations and challenges in blockchain have been articulately underlined by many researches [28][17] and articles [19][29], such as scalability, higher compute power, and storage. Nevertheless, many researchers have proposed solutions to overcome some of the challenges of blockchain for IoT. The authors Dorri et al. [30][31] have proposed solutions for IoTs based on blockchain and introduced a smart home based on blockchain. Albeit the proposed model still had some of the existing limitations. Subsequently, they proposed another optimized blockchain solution for IoT. The architecture of the proposed model is secure and suitable for IoTs, although, the additional overhead of keeping local (home) blockchain, overlay network nodes, and broadcasting between home IoTs still does not solve the network and communication problem.

C. Service composition in IoTs using blockchain

As mentioned earlier, many researchers have proposed solutions to overcome some of the challenges of blockchain for IoT. Although, none of the above studies have been performed under a service composition scheme using blockchain. The authors Ridhawi et al. [32] have proposed a blockchain-based composition solution. In their research, they have proposed a complex multimedia service delivery solution to cloud subscribers involving software defined networks (SDN), fog & edge computing, and reinforcement learning techniques for creating dynamic composition paths. The research exhibits that how blockchain can be used for the composition of complex services. Though the research has promising results, we argue that the use of blockchain limits the scalability of the proposed solution (as mentioned earlier [28][17][19][29]) and the proposed model does not directly involve computation on IoTs, rather on fog and edge nodes.

D. Service composition in IoTs using Tangle

With the idea of having secured immutable data, overcoming some of the shortcomings of blockchain for IoTs, a new technology IOTA Tangle [20] was proposed. The DAG-based blockchain structure for DLT is not new as it's been explored by many researchers [33][34]. The authors Živi et al. [35] have briefly explained the full potential of tangle architecture, the benefit of a tangle over blockchain, and the future of tangle. Similar studies have been performed by Wang et al. [36], Zander et al. [37] on the merits and demerits of the tangle.

Similar to the blockchain, each block in the tangle consists of transactions, however, a new block must reference two existing blocks attached to the tangle. Any client sending a transaction to tangle does not require storing the whole or partial tangle. This makes tangle a lightweight structure that can be run on any low compute and low-powered devices.

The figure 3 depicts a tangle structure. The red boxes in the figure imply new transactions arriving on a tangle edge called

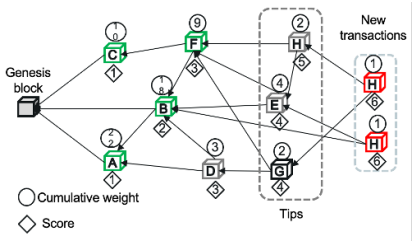


Fig. 3: IOTA Tangle

Tips. The grey boxes represent the tips of the tangle. The boxes in green are the confirmed transactions called milestones signed by a coordinator node. A milestone is a confirmed transaction verified and signed by a coordinator node and shared with everyone in the network as an indication that the transaction is valid, can be trusted, and now confirmed. The box in the grey color (denoted as D) is still an unconfirmed transaction because it is still not verified and signed by a coordinator node. The circle on a transaction box represents the cumulative weight of the transaction and the rhombus below the box represents a score of the transaction.

The use of PoW in the tangle differs from the blockchain. In tangle, PoW must be performed by a client by getting two tips of the tangle from a coordinator node and verifying it, to attach data to the tangle. This process is lightweight for IoTs (not like miners in the blockchain) as they have to just validate two transactions. The PoW mechanism also ensures that no client spamming a coordinator node.

The coordinator node is one of the most important parts of the tangle network. A coordinator node is responsible for holding full tangle data, selecting two tips on behalf of clients for them to verify, and validating tangle by running Tip's selection algorithm to ensure that the Tangle is growing in an honest direction. Also, a coordinator node is responsible for creating milestones.

One of the important features introduced with tangle is Masked Authenticated Messaging (MAM) [38]. It's a communication protocol that provides a mechanism to securely share and stream data (encrypted data) between clients on the tangle network. The MAM protocol uses channels to create secure communication between clients on top of the tangle network. Any clients subscribed to a channel (created using a private/shared key and the root of a Markle tree) can read the data from the channel. MAM supports 3 types of message mode namely public, restricted, and private. In private mode, the data published to a channel can only be read (decrypt data) and written (encrypted data) by a client if it has the secret key.

As tangle provides great benefits over blockchain and better suited for service composition, we propose a model utilizing tangle for service composition scheme in IoT and evaluate its feasibility using a smart home case study. The section III describes the components of proposed model and processing in our model is explained in section IV.

III. Components of proposed model

This section discusses the core components of the smart home model based on tangle as captured in figure 7.

A. Smart home devices (IoTs):

Nowadays, every household uses many electronic devices for accessibility. These devices are used for controlling a home environment such as lighting, temperature, lock, security, entertainment systems, etc. Generally, these devices are integrated with a home controller device for home automation which in turn controls these devices. When these devices are connected to the Internet, they are referred to as smart home IoTs. There are a plethora of IoTs present in the market for each home's needs and accessibility. Below are a few examples of smart home devices present in the market and used in our smart home model (along with acronyms used).

- Surveillance Camera: Capable of capturing image, facial recognition and sending message.
  acronyms: FC1 = Front Camera1, BC1 = Back Camera1
- Motion sensors: Capable of detecting moving object through heat & microwave and sending alert message with position, velocity, and acceleration.
  acronyms: FMD = Front motion detector, BMD = Back motion detector
- Door camera: Same capability as Surveillance camera.
  acronyms: DC = Door Camera
- Door Lock: Capable of capturing fingerprint/password/pin and sending message. acronyms: DL = Door Lock
- Ultrasound sensors: Capable of detecting objects through ultrasound waves and sending alert message with sound and distance. acronyms: USS1 = Ultrasound sensor 1, USS2 = Ultrasound sensor 2
- Vibration Sensor: Capable of detecting vibrations or tilt in any direction for windows. acronyms: VS1 = vibration sensor1, VS2 = vibration sensor2
- Smart TV: Capable of switching ON/OFF and receiving inputs. acronyms: TV1 = Smart TV1
- Smart Light/Bulb: Capable of switching ON/OFF and receiving inputs. acronyms: SL1 = Smart light1
- Thermostat: Capable of setting temperature based on readings and inputs. acronyms: TS1 = Thermostat1
- Water heater: Capable of heating water based on inputs. acronyms: WH = Water heater
- Home Controller: A home controller device (usually a higher compute power computer/device for computation and decision). acronyms: HC = Home Controller

B. Transaction:

A transaction in a smart home tangle is a request to perform some activity. The activity can be adding a device/IoTs to tangle, requesting and approving permission, or sharing data with other IoT. Therefore, based on activities, a transaction is categorized into 4 categories. The first transaction type is "add device" in which any device/IoTs are added to the tangle. The second transaction type is "request permission" in which IoTs request access to other home IoTs. To approve this request, a permission approval request is generated. And the last and final transaction type is "data" which is used for sharing data among IoTs.

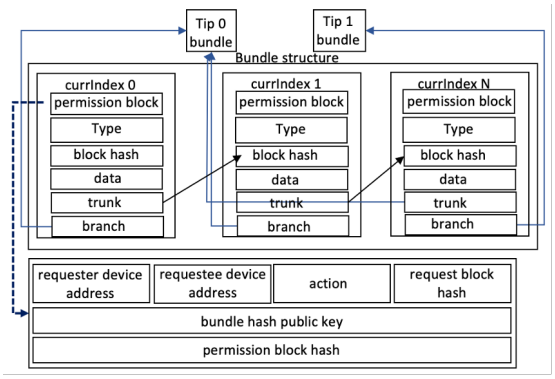


Fig. 4: Transaction bundle structure

The figure 4 depicts the transaction bundle structure. In the bundle, we have many transaction blocks which depend on the number of request permission or data share by an IoTs. A block in a transaction bundle is similar to a Tangle bundle but with an addition of a permission block field. A current index indicates the block number in the transaction block. The permission block field is described in III-C in details. The figure depicts the connectivity in a bundle and other fields. The data field in a block is encrypted using a shared key (III-E) to securely exchange data among home IoTs.

C. IoTs access and permissions:

It is important to restrict smart IoTs to communicate with each other based on permission. In the smart home tangle, permission to communicate with home IoTs are enclosed in the permission block field as shown in figure 5. The permission block consists of 6 parameters. The first 4 fields provide information on permission requests. The first parameter is "request device address" which contains the address of IoTs who is requesting permission. The "requestee device address" contains the address of devices to which this IoT would like to communicate. The "action" field indicates the action requested which can be either "allow" or "deny". The last parameter is the "request block hash" which refers to a permission request transaction bundle in a tangle. The field "bundle hash public key" contains the reference of an IoTs public key published in a tangle (III-D). The last parameter of the block is "permission block hash" which contains the signed hash of this permission request block.

Once an IoT is added to the tangle (IV-B), it can request permission from the homeowner to communicate with other home IoTs. This is achieved through creating a transaction by populating all necessary information except "request block

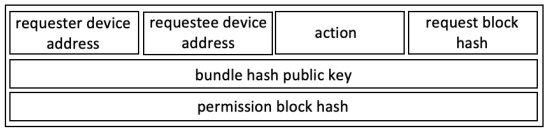


Fig. 5: Permission header structure in a transaction and sample example of request and approve permission.

hash". Since this is the permission request transaction, the field will be empty. An IoT can create multiple permission requests or combine all requests in one bundle with all the information.

Once a permission request transaction is created, it needs to be approved by the homeowner manually. Any permission request without homeowner approval is considered invalid. Once the homeowner approves the permission request, a permission approval transaction is created in the tangle. The "request block hash" of this approval transaction is populated with the permission request bundle hash, the "bundle hash public key" is populated with bundle hash of published homeowner private device public key (IV-B) in the tangle, and the "permission block hash" contains the singed hash of this approval (using a private key of a private device). This method ensures that all request permissions and approvals are verifiable by any other IoTs in the tangle to ensure the validity of permission requests, approval of permission request, associativity of an IoT with a home, and transaction.

D. IoTs identity and security:

In an untrusted environment, the identity of a device or its associativity with a home is quite challenging. To overcome this challenge, a private and a public key mechanism using the Diffie-Hellman [39] method is adapted.

First, a homeowner creates a private and public key using a private device (e.g., smartphone, personal laptop, or any device on which he/she wishes to be notified/alerted) in the Tangle network. The generated public key is considered as the home identifier. Similarly, to add a new IoT to the home, the owner must generate a private and public key for the IoT. The public key is the identifier for the IoT in the tangle network and home.

Any IoT added to the tangle can request for association with a home identifier. The approval of the request is manual, and each device needs to be approved by the homeowner to be trusted. If any registered IoT in the tangle is not associated with a home or a transaction request to associate with a home is not approved by a homeowner, the device or the request is considered invalid and will be removed from the tangle after a predefined time (configured). The benefit of this model is all IoTs will be associated with the homeowner public key, each home IoTs can validate the associativity of another IoTs, and any change or update permission will be broadcasted to all home IoTs for validation and future references.

E. Home keys:

To ensure secure communication between home IoTs, a home key is created and shared among all home IoTs. This key is used for creating channels (III-H) for secure data sharing and reading of secure data from the channel. An IoT who knows the key will be able to read or write encrypted data in the channel.

F. Tangle:

In the smart home model, a tangle running on a single home is considered a local tangle. If more than one home is connected with each other creating a tangle network, the tangle network will be a global tangle. In this case, all coordinator nodes (III-G) will behave similar to the IOTA Tangle coordinator (II-D) node and will continuously validate the tangle network and broadcast to each other.

G. Coordinator node:

In a single smart home, a computational powerful device needs to be set up as a coordinator node to hold the tangle and validate it. The coordinator node performs critical tasks in the tangle network as described in the II-D section. If a home occupant does not wish to host a coordinator node, they can connect to any neighboring peers hosting a global tangle network. In a global tangle network, the more the number of coordinator nodes connected, the more secure the tangle network will become.

H. Channels:

A channel in a smart home tangle is created using MAM protocol [38] to allow secure communication and data sharing among all home IoTs. To create a channel in private mode and publish data to the channel using MAM by a device, the device needs to have a private key. This private key is the home key (III-E) and is shared with all home IoTs when they are approved by the homeowner. To create a channel, an IoTs creates a Markel tree using the home key and the root of the market tree is the channel id. Whenever a new channel is created, the channel id is broadcasted to all home IoTs for listening and publishing data on this channel.

Using the home key and private mode of the MAM channel, any home IoTs can publish encrypted data to the next address in channel id or read encrypted data from the channel root. The channel is append only channel where the next address of the last block or message is always empty. Therefore, whenever a home IoTs publish the data to the channel, the data is added to the next address of the last block. Additionally, a channel can be created for a predefined time (configured) to share data. Any home IoTs can create a channel upon observing an activity, however, before creating a new channel, it will check if there exists a channel. To ensure the validity of the data in the channel, and additional validation can be performed by home IoTs.

IV. Smart home based on Tangle

In this section, we will describe our proposed model. First, we will look at the architecture of our model. Next, we will go over the initial setup required for setting up a smart home

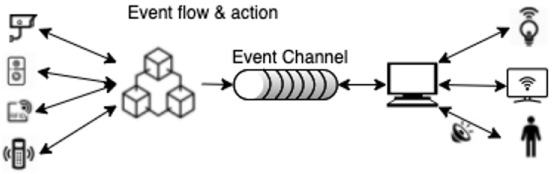


Fig. 6: A sample illustration of activity capture and service request with share data via channel on Tangle

based on Tangle. Thereafter, we will examine the transaction handling in our model. In the end, we will review the service compossibility achieved through Tangle in a smart home and study some sample workflows.

A. System architecture

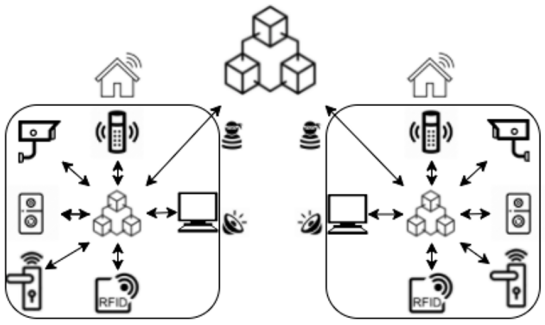


Fig. 7: A smart home setup with Tangle

B. Initial setup

In this section, we will describe the initial setup required in a smart home. First and foremost, a coordinator node (III-G) needs to be set up in a home. Once the coordinator node is up and running, the owner connects to the coordinator node via a private device (e.g., a smartphone or personal computer), gets an address (to receive alerts/notifications), and generates a private and a public key (III-D). The public key is stored in the tangle to the genesis block by creating a new block (III-B). The public key of this device will be used for verifying a home IoTs's associativity. Also, another private key (III-E) is created to be shared among future IoTs for creating channels (III-H) to securely communicate with each other.

Only after the first step is completed, a home IoTs can be added to the smart home tangle. To add an IoTs, an IoTs needs to connect to a tangle running on a coordinator node. The same steps are followed by an IoTs as owner private device for generating an address and creating a private and a public key. The public key is stored in the tangle and used for verifying the identity of the IoTs (III-D) and the data signed by it.

Now, to allow any newly added IoTs to communicate with each other, a new transaction needs to be issued by populating the permission headers field with details by following the permission block structure (III-C). The associativity of IoTs with a home is already discussed in section III-D and a transaction placement on tangle will be covered in section IV-C.

C. Transaction handling

Upon initial setup, a home IoTs can be added to the smart home tangle. To add a home IoT, the IoT needs to connect to the tangle. The same needs to be followed by an IoTs as owner private device for generating an address and creating a private and a public key. A transaction bundle is created by adding a public key in the data field and transaction type of "Add IoTs" and published to the tangle. The stored public key of an IoTs is used for verifying the identity of IoTs (III-D), its associativity with a home, and the data signed by it.

In the smart home model, a home IoT can communicate with any other home IoTs. Therefore, to allow any newly added IoTs to communicate with other home IoTs, a new transaction needs to be created by populating the permission headers field with details by following the permission block structure (III-C), transaction type as "Permission request" and requesting homeowner device for approval. The transaction is published to tangle and will be validated for its authenticity.

Any transaction block containing "Permission request" for policy addition/change/update need to be approved by the homeowner manually. This ensures the integrity of IoT permissions and the associativity of IoT with a home. A transaction containing a permission request without homeowner approval is considered invalid. Once the owner approves the permission request transaction, a new transaction block signed by the owner's private key is created and added to the tangle containing the reference to permission request transaction and transaction type of "Approval". Also, the transaction hash is broadcasted to all home IoTs of the home. Please refer to section III-C for permission transaction block creation & approval and III-D for IoTs associativity.

Now, all home IoTs associated with a home can communicate with each other based on permission. To ensure that the data sent from one home IoTs to another home IoTs is secure, a channel is created using MAM protocol (III-H). When a home IoTs wants to send data to another home IoTs, it creates a new transaction bundle block by following the transaction bundle structure and transaction type as "Data". The permission block field will be populated with the permissions (III-C) to indicate that IoT has permission to communicate with other IoTs. The data transaction bundle is published to the tangle and attached to the next address of the last block in the channel.

Whenever an IoTs wants to publish a transaction to the tangle, it must validate two tips of the tangle. The two tips are selected by the coordinator node by running the tips selection algorithm and sent to the IoTs for validation. The IoTs performs proof of work to validate the received tangle tips. The proof of work by IoTs ensures the validity of tips transaction and restrict IoTs to spam a coordinator node.

D. Service composition

The concept of IoTs services and their compositions has been already discussed in section II-A and II-D. Based on

that we will explore how service composition is achieved in our model.

In a smart home model, any home IoTs service can securely communicate with any other home IoTs and request for their service if the IoT has been granted permission. As a home IoTs detects activity, it can send the activity data securely to another IoTs service with appropriate service requests via channels for further actions. If a channel does not exist, it will create one, broadcast the channel id to all associated home IoTs, and sends the service request with the data to its permitted IoTs in the channel. On receiving the channel id, all home IoTs start monitoring the channel and data. The IoTs (who just received a service request and data from another IoTs) reads the data and tries to observe an activity. If it observes any activity, it sends further service requests and data to its permitted devices. If an IoTs who did not receive any service request and data, however still observers an activity, it requests for services from its permitted IoTs and includes its data in the request.

Any activity happening at the current moment and the appropriate IoTs observing the activity and requesting services from other home IoTs in the channel, the channel behaves as an event occurrence channel. Since all home IoTs are monitoring the channel, they can act or perform an action based on the data in the channel. As a home IoTs receives multiple service requests and perform actions based on the requests and data, it is an example of service compossibility in the smart home model. Figure 8 illustrates an example of events occurrence workflow when a home occupant enters the house. Based on this example event, we know that there can be many such service composition scenarios that can occur in the smart home. Hence in section IV-E, we will explore a few more examples of similar event capturing and service composition workflows.

E. Workflow example

1) Home occupant A entering home: When a home occupant enters a home, there are many activities observed and captured by different IoTs. The figure 8 highlights this service composition workflow.

As a home occupant moves towards the home and door, the FMD detects the presence of a moving object or person. The FMD request to monitor the activity to all permitted home IoTs (in this case home front camera1, door camera, and door lock) by sending a monitor service request and the captured data.

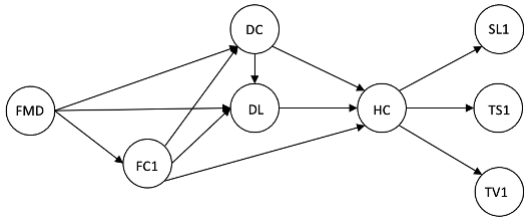


Fig. 8: Home occupant entering home events workflow

The IoTs who received the monitor request look for activity (person's activity) based on data and start monitoring. If FC1 observes the activity, it authenticates the person and requests further authentication to DC and DL. Similarly, as DC observes an activity and authenticates the person, it requests further services from DL and HC. Based on all the data and requests, either HC can send an open door request to DL, or DL itself can open the door automatically or wait for the person to provide further authentication (fingerprints/password/pin).

Based on data from all these home IoTs, an HC can find out if the person is authorized to enter the house. If the person is a home occupant, HC can further send a request to other home IoTs. In this case, the first service request is sent to the smart light. The temperature request is sent to the thermostat. And finally, the last service request is made to smart TV/Music system. The HC can request more services based on the current daytime and the configuration in a smart home model. However, if the person is not authorized to enter the house, and trying to forcefully enter the house, HC can take actions based on the data received from all IoTs and send an alert message to the home occupant with all evidence data and request an alert service from smart alarm to go off and ring.

2) Home occupant leaving home: As the name suggests, this workflow captures home occupant leaving events. As an occupant moves towards the front door, USS1 captures the activity, creates a channel, and sends an alert service request to HC. Also, as the occupant moves out of the home by closing the door, the DL and DC both capture the events. DC authenticates the occupant and sends a monitor service request to FC1, FMD, and HC. Similar to DC, FC1 captures the image, authenticates the occupant, and sends a monitor request to FMD and HC. The FMD detects the moving activity and sends a monitor service request back to FC1, DC, and HC along with captured data (position, velocity, etc.). The data from FMD will indicate the occupant is moving away from the home.

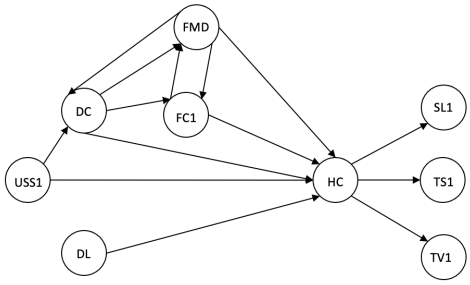


Fig. 9: Home occupant leaving home events workflow

As all home IoTs are publishing the data in a channel and monitoring the channel, anyone can take further actions based on the data in the channel. In this case, the HC can request for lock service from DL, or DL can itself take a decision

based on data in the channel. Additionally, since the occupant has left the home, HC can request all other home IoTs to turn OFF to save energy consumption. In this case, it requests smart light SL1, TV1 to turn OFF and the thermostat to turn OFF AC and consume less power.

3) Home occupant B entering home: The workflow scenario is similar to the IV-E1 except it is authorizing another occupant of the house. If another occupant has been registered in the home IoTs, the events will be observed in the same fashion as IV-E1 and the occupant will be authenticated and allowed. However, based on the configuration, this time HC can take different actions while requesting services from other IoTs. For example, for this occupant, if it configured to request smart light to turn ON and change the light color to light blue and turn ON smart music system and play favorite music or specific genre songs.

4) Forceful entering house through front window break-in: As the workflow name suggests, in this case, a burglar or unknown person trying to enter the house by breaking in through the front window and entering the house. As the person moves towards the house, his activity will be observed by FMD or FC1. If FDM detects any motion, it can request FC1, DC, and DL for monitor service. Similarly, FC1 will capture the image, validate authorization (unauthorized in this case), and request an additional authorization service request from DC and DL.

However, let's assume the person knew about the FMD and FC1 and avoided both in the way to reach the front window. In this situation, neither FMD nor FC1 will observe the activity and hence no further authentication request to DC and DL. However, as the person tries to break through the window or impact in any way, the VS1 device will sense the vibration or tilt in the window and will immediately send an alert request to the HC. Similarly, as the person moves into the house, the USS1 will observe the presence and will send an alert service request to the HC.

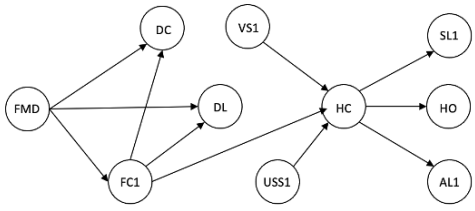


Fig. 10: A break-in home events workflow in front of a home

Now, as the HC receives these requests and data from VS1 and USS1, it immediately makes few service requests. First, it will request the smart light to turn ON. Secondly, it will send an alert message to the home occupant and finally, it will raise an alert request to the smart alarm to go off and ring. (The HC can take an additional step to call home occupant or any security personal or service). Therefore, in this scenario, all the observed data will be sent to the home occupant as evidence and force the unauthorized person to panic and leave immediately.

5) Forceful entering house through back window break-in:

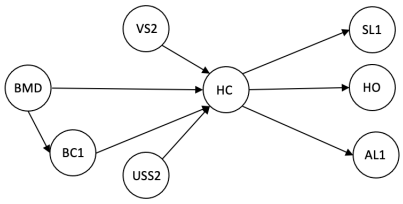


Fig. 11: A back window break-in home events workflow

The workflow scenario is similar to IV-E4 expect an unauthorized person trying to break in through the back-side window of the home. However, in this scenario, the activities will be observed by BMD, BC1 or VS2, and USS2 as the person break-in and enters the home. Based on the service request from all these IoTs and the data, HC will perform the same action as performed for the IV-E4 section.

6) Home owner presence and activity: An HC can be configured to request other home IoTs to perform certain actions depending upon the presence of an occupant and the time of the day. let's say an occupant gets up at night and move towards the hall or kitchen. The USS1 will detects the activity and captures the data. At the same time, it creates a channel and broadcasts the channel id to all home IoTs, and requests an alert service from HC in the channel. As HC already aware of the presence of the occupant, it can request SL1 for its light service to turn ON and set the light color to mild white. Similarly, as no activity is detected by USS1, it will send no activity data to HC as information in the channel. Accordingly, the HC can decide to request other IoTs such as SL1, TV1, or any other IoTs for their service and turn OFF to save energy consumption.

As mentioned earlier, there can be many such examples of service compossibility that can occur in the smart home model. It is not possible to include all such workflow in this study, however, the important workflows are already been covered. This indicates that the service compossibility can be achieved in the smart home model using tangle which is the primary objective of this study.

V. Results and performance evaluation

A. Evaluations and Results

As explained in section IV-E, different service composition workflows can occur in a smart home. Therefore, we described the results of each workflow below. Please refer to III-A for the naming convention used in the results. Additionally, only the IoTs mentioned in the result tables are the participants in a workflow. The verb "ACTION" and "NO ACTION" is used to describe the service composition occurrence in our workflow.

The verb ”ACTION” means, service compositions from participating IoTs occurred and an HC has further composed the data and requested additional services. The verb ”NO ACTION” means, service compositions from participating IoTs occurred, however, HC has not composed the data from a channel and hence not requested any additional services. The ”Time” (in seconds) column indicates the average time taken from the time when an IoT published its data in a channel to the time it taken HC to make service requests.

1) Home occupant ”A” entering home: When someone enters a home, there are many activities observed and events captured by different home IoTs. therefore, we simulated the most common scenarios when someone enters a home. We simulated for 40 times which includes an authorized person (in our case occupant A or B) entry 20 times and an unauthorized person trying to enter the home 20 times. The table I highlights all the events simulated for this workflow.

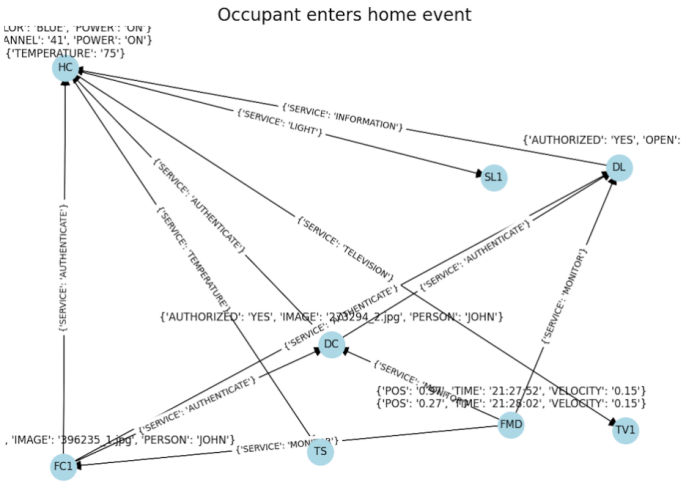


Fig. 12: service compositions when occupant ”A” enters home event

The first row of the table indicates that all IoTs observed the events and requested services from permitted home IoTs by publishing data in a channel. For the first 20 authorized entry simulations, HC has further requested services from SL1, TV1, and TS. The figure 12 highlights the service compositions events in this workflow. For the other 20 unauthorized entry simulations, HC has requested services from SL1 and alerted the homeowner with unauthorized activity.

The rows from second to six highlight the events in which one of the IoTs did not observe activity for all 40 runs. Even in the absence of data from one of the IoTs for the first 20 authorized entries, HC was able to make the right decision by requesting SL1, TV1, and TS for their services. For the other 20 unauthorized activity, again, HC was able to request service from SL1 and alerted the homeowner with unauthorized activity.

For the remaining entries from seven to sixteen, we assumed two IoTs did not observe activity and simulated the events.

Seq	FMD	FC1	DC	DL	USS1	HC	SL1	TV1	TS	HO	Time (secs)
1	40	40	40	40	40	ACTION*	40	13	20	20	0.575
2	-	40	40	40	40	ACTION*	40	10	20	20	0.638
3	40	-	40	40	40	ACTION*	40	14	20	20	0.644
4	40	40	-	40	40	ACTION*	40	12	20	20	0.671
5	40	40	40	-	40	ACTION*	40	10	20	20	0.593
6	40	40	40	40	-	ACTION*	40	12	20	20	0.542
7	-	-	40	40	40	ACTION*	40	10	20	20	0.515
8	-	40	-	40	40	ACTION*	40	13	20	20	0.672
9	-	40	40	-	40	ACTION*	40	9	20	20	0.579
10	-	40	40	40	-	ACTION*	40	12	20	20	0.658
11	40	-	-	40	40	NO ACTION+	-	-	-	-	0.547
12	40	-	40	-	40	NO ACTION+	-	-	-	-	0.489
13	40	-	40	40	-	ACTION*	40	14	20	20	0.516
14	40	40	-	-	40	NO ACTION+	-	-	-	-	0.497
15	40	40	-	40	-	ACTION*	40	10	20	20	0.491
16	40	40	40	-	-	ACTION*	40	13	20	20	0.512

* Composition of FMD, FC1, DC, DL, and USS1. HC requests SL1, TV1, and TS to turn ON, set light color, set channel, adjust the thermostat. Also, HC notifies the home occupant when an unauthorized person tries to enter the home.

+ Composition of FMD, FC1, DC, DL, and USS1 but no further requests (composition).

!! The same service compositions occur when occupant ”B” enters home except for the different configuration of service requests to SL1, TV1, and TS.

TABLE I: Service compositions scenarios when occupant ”A” enters home

In this scenario, for the first 20 authorized entries, HC was able to request additional services in some cases when it was sure of an authorized entry. However, for other cases, it was not able to request additional services. The event captured in row eleven is an interesting event. Even though the DL has indicated the authorized person’s activity (by entering fingerprint/password/pin), however, HC wasn’t sure as no other IoT data corroborated with DL data. For the remaining 20 unauthorized activities, HC was able to request appropriate service and notified the homeowner.

2) Home occupant ”B” entering home: The same events workflow has occurred when occupant “B” enters the home. The figure 13 highlights this service compositions event. When occupant “B” enters the home, a different set of configurations of service requests sent to SL1, TV1, and TS. Apart from it, all the events had the same behavior except for service time which was similar to the time in table I.

3) Home occupant leaving home: We simulated this workflow 20 times with different events. The table II describes all the events simulated in this workflow. The data in the first row indicates that all participating IoTs have captured

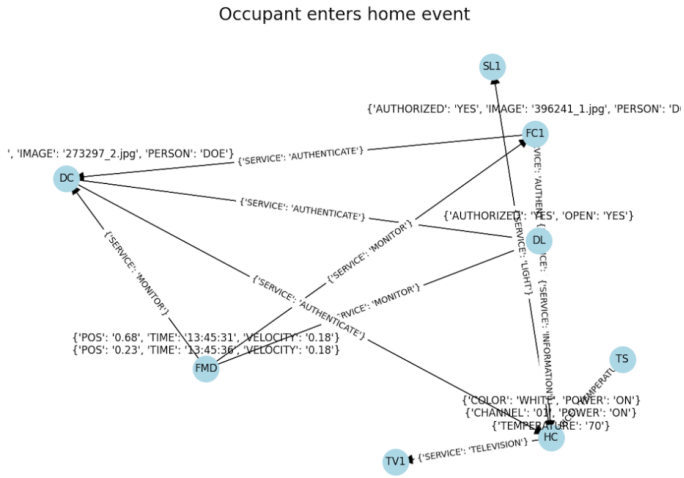


Fig. 13: Service compositions when occupant "B" enters home event

the occupant leaving event and therefore HC has requested SL1, TV1, and TS for their services to turn OFF and adjust the home temperature. All the events and service compositions of this event are highlighted in the figure 14.

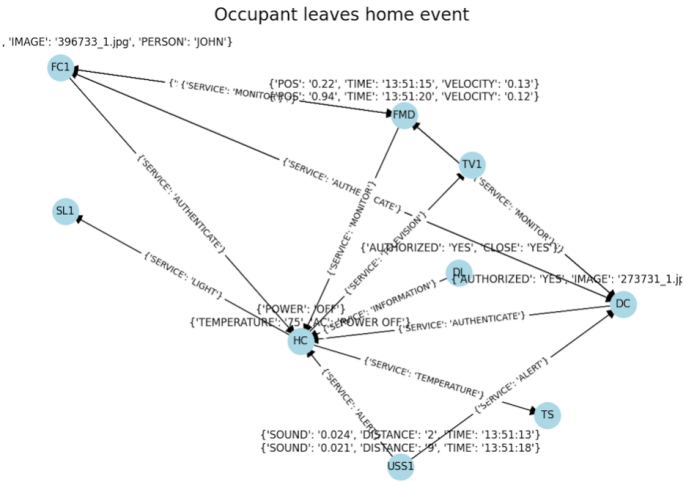


Fig. 14: Service compositions when an occupant leaves home event

Similar to the home occupant entering the home scenario, the rows from second to six describes the events where one of the IoTs did capture any activity and the remaining entries in which two home IoTs did not capture any activity. The service composition took place in all the events in this workflow and HC has further requested services from SL1, TV1, and TS except for entries eleven and twelve. In these two cases, HC was not able to formulate a decision for additional services because the information from other home IoTs was not enough to make service requests.

Also, notice that in rows three, seven, and thirteen, HC has additionally requested DL for its service to lock the door as the occupant might have missed locking the door, and hence DL did not publish lock event data in the channel.

Seq	USS1	DL	DC	FC1	FMD	HC	SL1	TV1	TS	DL	Time (secs)
1	20	20	20	20	20	ACTION*	20	20	20	-	0.676
2	-	20	20	20	20	ACTION*	20	20	20	-	0.521
3	20	-	20	20	20	ACTION*	20	20	20	20	0.473
4	20	20	-	-	20	ACTION*	20	20	20	-	0.754
5	20	20	20	-	20	ACTION*	20	20	20	-	0.411
6	20	20	20	20	-	ACTION*	20	20	20	-	0.781
7	-	-	20	20	20	ACTION*	20	20	20	20	0.539
8	-	20	-	20	20	ACTION*	20	20	20	-	0.490
9	-	20	20	-	20	ACTION*	20	20	20	-	0.381
10	-	20	20	20	-	ACTION*	20	20	20	-	0.487
11	20	-	-	20	20	NO ACTION+	-	-	-	-	0.397
12	20	-	20	-	20	NO ACTION+	-	-	-	-	0.433
13	20	-	20	20	-	ACTION*	20	20	20	20	0.464
14	20	20	-	-	20	ACTION*	20	20	20	-	0.492
15	20	20	-	20	-	ACTION*	20	20	20	-	0.463
16	20	20	20	-	-	ACTION*	20	20	20	-	0.434

* Composition of USS1, DL, DC, FC1, and FMD. HC requests SL1, TV1, and TS to turn OFF. Also, HC request DL to lock the door in some scenarios.
+ Composition of USS1, DL, DC, FC1, and FMD but no further requests (composition).
!! The same service compositions occur when occupant "B" leaves home.

TABLE II: Service compositions scenarios when occupant "A" leaves home

4) Forceful entering house through front window break-in: In this workflow, we simulated 20 times for an unauthorized person trying to break in through the front window and entering the home. The table III highlights all the events simulated for this workflow. The first row of the table indicates that all IoTs observed the events and requested services from permitted home IoTs, and HC has further requested service from SL1 and alerted the homeowner with break-in activity. The figure 15 highlights the service compositions events.

The rows from second to five highlight the events in which one of the IoTs did not observe an activity. In the absence of activity data from one of the IoTs, HC was able to request service from SL1 and alerted the homeowner with break-in activity except for one case (row fourth). In the row fourth event, HC was not able to request further service since there was no front window break-in activity.

For the remaining entries events in the table when two IoTs did not observe an activity, the HC was not able to make a further service request and alert the homeowner except for the

Seq	FMD	FC1	VS1	USS1	HC	SL1	HO	Time (secs)
1	20	20	20	20	ACTION*	20	20	0.538
2	-	20	20	20	ACTION*	20	20	0.531
3	20	-	20	20	ACTION*	20	20	0.588
4	20	20	-	20	NO ACTION+	-	-	0.496
5	20	20	20	-	ACTION*	20	20	0.481
6	-	-	20	20	ACTION*	20	20	0.512
7	-	20	-	20	NO ACTION+	-	-	0.468
8	-	20	20	-	ACTION*	20	20	0.471
9	20	-	-	20	NO ACTION+	-	-	0.541
10	20	-	20	-	ACTION*	20	20	0.419
11	20	20	-	-	NO ACTION+	-	-	0.503

* Composition of FMD, FC1, VS1 and USS1 on HC and requests to SL1 to turn ON and notifies home occupant.
+ Composition of FMD, FC1, VS1 and USS1 on HC but no further requests (composition).

TABLE III: Service compositions scenarios when front window break-in detected

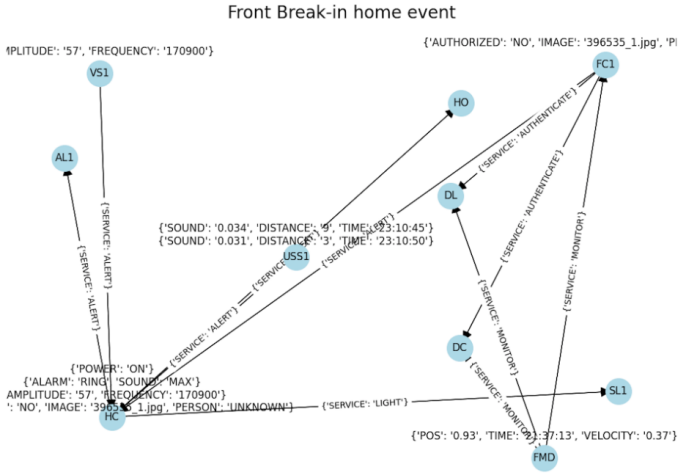


Fig. 15: Service compositions when front window break-in detected event

event in row tenth where front window break-in activity was detected.

5) Forceful entering house through back window break-in: The same events workflow occurred when we tried to simulate an unauthorized person break-in and entering through the back window of the home. In this case, different IoTs observed all the events than V-A4 (BC1 instead of FC1, VS2 instead of VS1, and USS2 instead of USS1), however, their function and behavior of capturing activity are the same. Also, the service time were similar to table III. The figure 16 highlights this service compositions event.

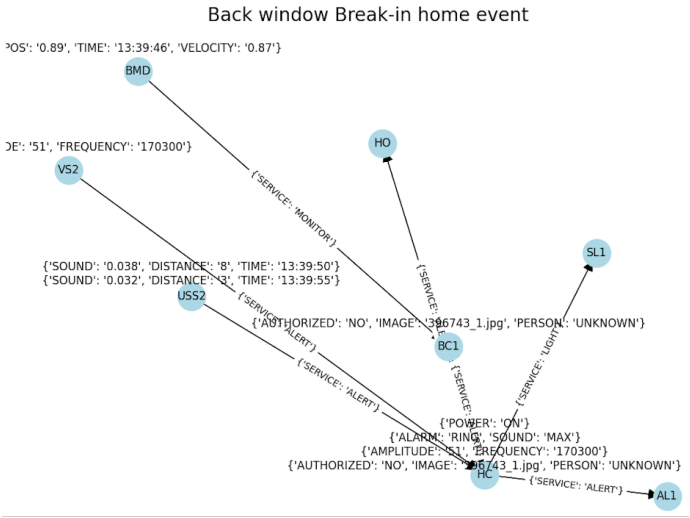


Fig. 16: Service compositions when back window break-in detected event

Seq	USS1	USS2	HC	SL1	TV1	WH	time (seconds)
1	20	20	ACTION*	20	12	3	0.358
2	-	20	ACTION*	20	14	3	0.305
3	20	-	ACTION*	20	11	2	0.378
4	-	-	ACTION+	20	20	20	0.292

* Composition of USS1 and USS2 on HC and requests to SL1, TV1, and WH to turn ON, set color, set channel, and heat water.
+ Composition of USS1 and USS2 on HC and requests to SL1, TV1, and WH to turn OFF.

TABLE IV: Service compositions scenarios when an occupant at home

6) Home occupant presence and activity: We had only two IoTs in this workflow which notifies the presence and activity of a person (in this case home occupant). Again, we simulated 20 runs for an occupant presence and activity. The table IV outlines the events simulated for this workflow. The first row indicates that both IoTs observe the presence and some activity. Hence HC has further requested services from SL1, TV1, and WH. The figure 17 outlines the service compositions events of this workflow. The services requested from WH only at a specific time (6 AM-9 AM and 6 PM-8 PM) of the day.

The rows second and third represents that only one of the IoTs observed the presence and hence HC has requested services. The row fourth indicates that none of the IoTs observed only presence and activity and therefore HC has requested services from other IoTs to turn OFF.

Along with simulating for all the service composition scenarios and workflow events, we had captured the CPU and memory consumption of IoTs running our code. Since we had simulated all workflow scenarios separately at different times, we noted the maximum CPU and memory consumption of

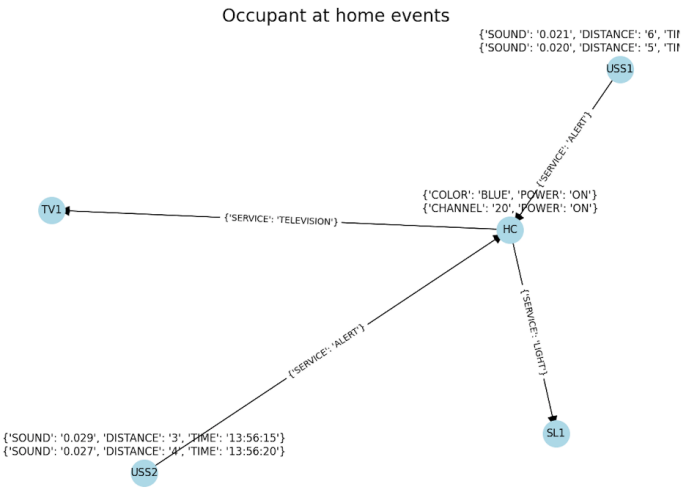


Fig. 17: Service compositions when Occupant "A" at home event

each scenario and averaged all the readings. Table V highlights the average maximum CPU and memory consumption of our simulation. Table V, indicates that the CPU usage in all home IoTs are comparatively less and memory consumption is minimal. However, the coordinator node has higher memory consumption and high memory utilization. This is expected as a coordinator node stores the tangle and runs the tips selection algorithm.

Name	CPU (%)	Memory	Memory utilized	Memory utilized (%)
COO*	67.62	4 GB	104.64 MB	26.16
FMD	32.98	1 GB	26.41 MB	2.58
HC	57.31	2 GB	45.70 MB	2.23
FC1	55.47	1 GB	39.87 MB	3.89
DC	59.55	1 GB	30.72 MB	3.00
DL	56.31	1 GB	27.40 MB	2.68
USS1	30.04	1 GB	22.81 MB	2.23
USS2	33.28	1 GB	29.94 MB	2.91
BMD	43.82	1 GB	33.45 MB	2.43
BC1	61.32	1 GB	53.23 MB	5.13
VS1	41.52	1 GB	33.40 MB	3.23
VS2	39.43	1 GB	37.47 MB	3.63

* A coordinator node running tangle

TABLE V: Average CPU and memory usage of home IoTs in service composition scenarios

In our work, we have simulated different common events workflow which occur normally. During our experiment, we had sent 4460 transactions covering all the above scenarios. However, there were some event scenarios where further service compositions did not occur on HC as expected. Table VI indicates the accuracy matrices for our experiment. The table delineates the actual service compositions and all executed scenarios service compositions. Additionally, it highlights how many times the executed service composition scenarios did not occur as expected.

Actual/In execution	Service compositions by controller in execution	NO service composition by controller in execution
Actual service compositions by controller	3180	360
Actual NO service composition by controller	440	480

TABLE VI: Accuracy metrics for all service compositions in our experiment

VI. Considerations

To add any new device to tangle to communicate with other home IoTs, IoTs request permission must be approved by a homeowner manually. Any permission request without approval transaction reference is considered invalid and removed from tangle (III-C). Hence, due to this, no exploit can be performed by an adversary to add any malfunctioning IoTs to a home. In addition to that, the permission block provides additional security for communication between IoTs. An IoT request permission to communicate with other home IoTs by including permission request in the permission block for each IoTs communication. Therefore, if an IoTs has not been permitted to access another home IoTs, the IoTs cannot communicate with the other home IoTs, and hence any such transaction will be rejected.

Now we will consider a scenario, where a home IoT device is hacked and an adversary trying to manipulate data with it. However, in our model, an HC is reading through a channel created, reading all the observed data, and making decisions based on that. Therefore, HC will make the right decision accordingly. Also, since the malicious data would not corroborate with other observed data in the channel, the HC can notify a homeowner of the malicious behaviors. This will notify a homeowner about any malicious activity of an IoTs. However, if more than one home IoTs has been hacked, in some cases, the HC might not be able to take the right decision and hence will only notify the homeowner about the activity.

What if our HC is hacked since it makes the decision and service composition? The HC is running the decision-making capabilities in the home. This model can be extended to allow running the decision-making algorithm on home IoTs directly along with their normal function.

As HC just requests services from other home IoTs, any IoTs can request the same with an additional capability of broadcast. In this case, once a service request is published to tangle, and IoTs can broadcast the bundle id to other home IoTs to indicate that it has requested service. All home IoTs can validate the request and accordingly service requests will be performed. This model will give an additional benefit of not trusting any home IoTs, validating all service requests, and deciding based on all the observed data in a channel.

Apart from the above security concerns, the tangle can be attacked with a parasite chain attack [40]. However, as

mentioned in the authors [20], running an appropriate tips selection algorithm such as the Monte Carlo Markov Chain algorithm on the coordinator node will ensure the tangle will grow in an honest direction. Additionally, the detection of such parasite chain can be achieved by the proposed model in [41] and make an appropriate decision. A coordinator node can be hosted by an adversary and hence can influence the tangle in our model. However, we have proposed a solution to handle this scenario in the future work section.

VII. Future Work

One of the important notes on a coordinator node is that it is the critical part of the tangle network. The coordinator node is responsible for holding the tangle, selecting two tips by running the tips selection algorithm and validating the tangle, and sending the tips to IoTs to validate. Also, the coordinator node is responsible for creating milestones to be shared among other coordinators nodes or other nodes hosting tangle and informing them that all transactions till this milestone are have been verified and signed by it. Because of all these responsibilities and critical work of the tangle network, the majority of the coordinator nodes are hosted by the IOTA foundation to ensure the tangle's security and authenticity. Due to this reason, the tangle is not considered fully decentralized.

Since our model also relies on the coordinator node for its responsibilities, any home coordinator can try and employ higher compute power 34% [42] of the tangle network to attack or change the tangle direction. This becomes a challenging and bottleneck problem for our model as well. Many researchers have been trying to find a solution for such an uncertain problem. Recently, IOTA Foundation has proposed to remove the coordinator node completely from the tangle network and introducing a coordicide node. The coordicide node is similar to the coordinator node, however, it will be maintained by all volunteers and it's running the Fast Probabilistic Consensus in Byzantine Infrastructure (FPC-BI) [43] algorithm based on a voting system to build trust between coordicide nodes. This will ensure the integrity of the coordicide node and the authenticity of the tangle as a byproduct.

In future work, we can employ the coordicide node features and FPC-BI algorithm to ensure our smart home tangle security and integrity. Also, we can strengthen the confidence and trust in our coordicide node by additionally adding Proof of Authority [44] or Proof of Trust [45] based mechanism.

VIII. Conclusion

In this paper, we have discussed the service compossibility in IoTs and how blockchain-based technology solves security and privacy issues associated with IoT. We also examined that blockchain-based solutions for IoTs are still not suitable due to high computation and scalability issues. We explored an alternative of blockchain namely tangle technology and its benefit over blockchain for IoT. Then, we explored and introduced a model to exhibit the feasibility of service composition in IoTs based on tangle technology using a smart home case study.

In the paper, we outline our proposed model with a smart home case study and the core components of our model. Additionally, we described how the placement of transactions is handled and how the compossibility of home IoTs services achieved in a smart home with a few smart home workflow scenarios. Thereafter, we presented the results of our simulation for different home IoTs service compossibility. Apart from the results, we also outlined some considerations and security analysis in our model. Based on our simulation results, we demonstrated that the composition of IoTs services can be achieved through the tangle with low computation and storage with a little overhead of broadcast between home IoTs. However, we argue that the extra overhead of broadcasting has a huge benefit of high integrity, security, and privacy in a smart home.

To the best of our knowledge, this is the first study utilizing the composition of IoTs services using tangle technology with a smart home use case. In the future, we can explore the feasibility of our proposed work in the tangle described in the future work section.

Acknowledgment

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